Consider the following transitions:

Transitions	$m \rightarrow n$
1	$3 \rightarrow 2$
2	$2 \rightarrow 1$
3	$3 \rightarrow 1$

Show that the frequency of these transitions obey sum rule (which is known as Ritz combination principle)

[Ans: $\nu_{3\rightarrow2} + \nu_{2\rightarrow1} = \nu_{3\rightarrow1}$]

3. (a) A hydrogen atom is excited by radiation of wavelength 97.5 nm. Find the principal quantum number of the excited state.

(b) Show that the total number of lines in emission spectrum is $\frac{n(n-1)}{2}$. Compute the total number of possible lines in emission spectrum as given in(a).

[Ans: (a) n =4 (b) 6 possible transitions]

4. Calculate the radius of the earth if the density of the earth is equal to the density of the nucleus.[mass of earth $5.97\times10^{24} kg$].

[Ans:180 m]

5. Calculate the mass defect and the binding energy per nucleon of the $^{108}_{47}Ag$ nucleus. [atomic mass of Ag = 107.905949]

Ans: $\begin{bmatrix} \Delta m = 0.990391u \text{ and} \\ \overline{B.E} = 8.5MeV \end{bmatrix}$

6. Half lives of two radioactive elements A and B are 20 minutes and 40 minutes respectively. Initially, the samples have equal number of nuclei. Calculate the ratio of decayed numbers of A and B nuclei after 80 minutes.

[Ans: 5:4]

7. On your birthday, you measure the activity of the sample ^{210}Bi which has a half-life of 5.01 days. The initial activity that you measure is $1\mu Ci$. (a) What is the approximate activity of the sample on your next birthday? Calculate (b) the decay constant (c) the mean life (d) initial number of atoms.

[Ans: (a) $10^{-22}\mu Ci$ (b) $1.6\times10^{-6}\ s^{-1}$

(c) 7.23days (d) 2.31×10^{10}]

8. Calculate the time required for 60% of a sample of radon undergo decay. Given $T_{1/2}$ of radon =3.8 days

[Ans: 5.022 days]

9. Assuming that energy released by the fission of a single $^{235}_{92}U$ nucleus is 200MeV, calculate the number of fissions per second required to produce 1 watt power.

[Ans: 3.125×10^{10}]

10. Show that the mass of radium ($^{226}_{88}Ra$) with an activity of 1 curie is almost a gram. Given $T_{1/2}$ =1600 years.

11. Charcol pieces of tree is found from an archeological site. The carbon-14 content of this charcol is only 17.5% that of equivalent sample of carbon from a living tree. What is the age of tree?

[Ans: 1.44×10^4 yr]

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